Paper Summary:
Direct3D: Scalable Image-to-3D Generation via 3D Latent Diffusion Transformer

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Citation

Shuang Wu, Youtian Lin, Feihu Zhang, Yifei Zeng, Jingxi Xu, Philip Torr, Xun Cao, Yao Yao. *Direct3D: Scalable Image-to-3D Generation via 3D Latent Diffusion Transformer.* Advances in Neural Information Processing Systems (NeurIPS) 37, 2024. `https://proceedings.neurips.cc/paper_files/paper/2024/hash/dc970c91c0a82c6e4cb3c4af7bff5388-Abstract-Conference.html`

1 Problem and Motivation

Generating 3D assets from a single image is difficult mainly because 3D training data (e.g., Objaverse-XL, ∼10M shapes) is far smaller and less diverse than 2D image datasets (e.g., LAION-5B, ∼5B images). To work around this, most prior image-to-3D pipelines are *indirect*: they first use a multi-view diffusion model to synthesize several 2D views of an object from one input image, and then reconstruct a 3D shape from those views using sparse-view reconstruction or Score Distillation Sampling (SDS). This two-stage design is slow and its output quality is capped by the fidelity of the intermediate multi-view images, often producing detail loss or reconstruction artifacts.

Direct3D instead trains a *native* 3D generative model that maps an image directly to a 3D shape in a single diffusion process, avoiding both multi-view diffusion and SDS optimization.

2 Method

The model has two components trained in two stages, inspired by Latent Diffusion Models (LDM): a 3D variational autoencoder (D3D-VAE) that compresses 3D shapes into a compact latent space, and a diffusion transformer (D3D-DiT) that generates new latents in that space, conditioned on an image.

2.1 D3D-VAE: Direct 3D Variational Autoencoder

- **Point-to-latent encoder.** A dense point cloud (with normals) is sampled from the surface of a 3D object and encoded, via cross-attention with a set of learnable tokens followed by self-attention layers, into an explicit *triplane* latent: three orthogonal 2D feature maps (XY, YZ, XZ planes) stacked together, rather than the unstructured 1D implicit latent used by prior work (e.g., Shap-E, Michelangelo).

- **Latent-to-triplane decoder.** A convolutional network upsamples the low-resolution latent triplane into high-resolution triplane feature maps.
- **Geometric mapping network.** An MLP predicts the occupancy of any queried 3D point from features interpolated off the triplane.
- **Semi-continuous surface sampling.** Instead of supervising occupancy with a hard 0/1 label (which causes unstable gradients near the surface) or with rendered-image losses (used by most prior VAEs), the authors supervise occupancy directly with a signed-distance-based value that is discrete far from the surface and continuous close to it:

$$o(x) = \begin{cases} 1, & \mathrm{sdf}(x) < -s \\ 0.5 - \dfrac{0.5 \cdot \mathrm{sdf}(x)}{s}, & -s \le \mathrm{sdf}(x) \le s \\ 0, & \mathrm{sdf}(x) > s \end{cases}$$

  with threshold $s = 1/512$. Training minimizes a binary cross-entropy occupancy loss plus a KL regularization term on the latent, i.e., $\mathcal{L}_{\text{D3D-VAE}} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}}$.

2.2 D3D-DiT: Image-Conditioned 3D Diffusion Transformer

Because a naive 2D U-Net would not let the three triplane feature maps exchange information, the authors use a Diffusion Transformer (DiT-XL/2 backbone, 28 blocks) operating on the flattened triplane latent, which better fuses positional information across planes. Image conditioning is injected at two levels in every DiT block:

- **Pixel-level alignment.** DINO-v2 (ViT-L/14) features of the input image are projected and concatenated with the noisy latent tokens, then fused via self-attention, to preserve fine structural/geometric detail.
- **Semantic-level alignment.** CLIP (ViT-L/14) features of the input image are fused via cross-attention, and the CLIP class token is added to the time embedding, to keep the generated shape semantically consistent with the image.

The model is trained as a standard denoising diffusion model (predicting noise ϵ added to the latent at timestep t), with 10% random conditioning dropout for classifier-free guidance at inference (1000 training steps, 50-step DDIM sampling at inference, guidance scale 7.5).

2.3 Training Data

Trained on a filtered 160K-shape subset of Objaverse (plus an internal dataset in the larger-scale variant), each shape normalized to a unit sphere. Conditioning images are 24 rendered views per shape (512×512, via Blender) augmented with 16 ControlNet depth-conditioned images per shape for better generalization to real/in-the-wild images.

3 Results

3.1 Image-to-3D (Google Scanned Objects benchmark, 30 shapes)

3.2 Text-to-3D and User Study

Using text-to-image models (Flux, Hunyuan-DiT) to first synthesize a conditioning image, Direct3D produces coherent, high-quality meshes where most baselines fail outright. A 46-person user study

Method	Chamfer Distance ↓	Volume IoU ↑	F-Score ↑
Shap-E	0.0585	0.2347	0.3474
Michelangelo	0.0441	0.1260	0.4371
One-2-3-45	0.0513	0.2868	0.3149
InstantMesh	0.0327	0.4105	0.5058
Direct3D (Objaverse only)	0.0296	0.4307	0.5356
Direct3D (+ internal data)	**0.0271**	**0.4323**	**0.5624**

Table 1: Direct3D outperforms all baselines on every metric; adding more training data further improves results, demonstrating scalability.

rating mesh quality and image consistency (1–5 scale) gave Direct3D scores of 4.41 (quality) and 4.35 (consistency), far above the next-best baselines (InstantMesh: 2.53 / 2.66).

3.3 Qualitative Findings

Compared to baselines, Direct3D avoids common failure modes such as holes/artifacts (Shap-E), semantic misalignment with the input image (Michelangelo), coarse geometry from single-view sparse reconstruction (One-2-3-45), and multi-view inconsistency such as merged or duplicated parts (InstantMesh). Generated meshes can also be textured afterward with an off-the-shelf method (SyncMVD).

4 Contributions

1. First native 3D generative model shown to scale to in-the-wild input images, without needing multi-view diffusion or SDS optimization.
2. D3D-VAE: an explicit triplane-latent 3D autoencoder supervised directly on decoded geometry (semi-continuous occupancy) rather than via differentiable rendering losses.
3. D3D-DiT: an image-conditioned diffusion transformer that fuses positional information across the triplane and combines pixel-level (DINO-v2) and semantic-level (CLIP) image conditioning.
4. State-of-the-art image-to-3D and text-to-3D generation quality and generalization, validated quantitatively (GSO benchmark) and via human evaluation.

5 Limitations

The method currently generates individual or multiple discrete objects and cannot yet generate large-scale 3D scenes; the authors note this as future work.

6 Take-away

Direct3D's central idea is to replace the common "generate multiple 2D views, then reconstruct in 3D" pipeline with a single native 3D latent diffusion model, made feasible by (1) an explicit, compact triplane latent trained with direct geometric (occupancy) supervision instead of rendering losses, and (2) a transformer-based diffusion architecture that injects both low-level (pixel) and high-level (semantic) image information at every layer. This yields faster, more consistent, and better-generalizing image/text-to-3D generation than prior multi-view- or implicit-latent-based approaches.